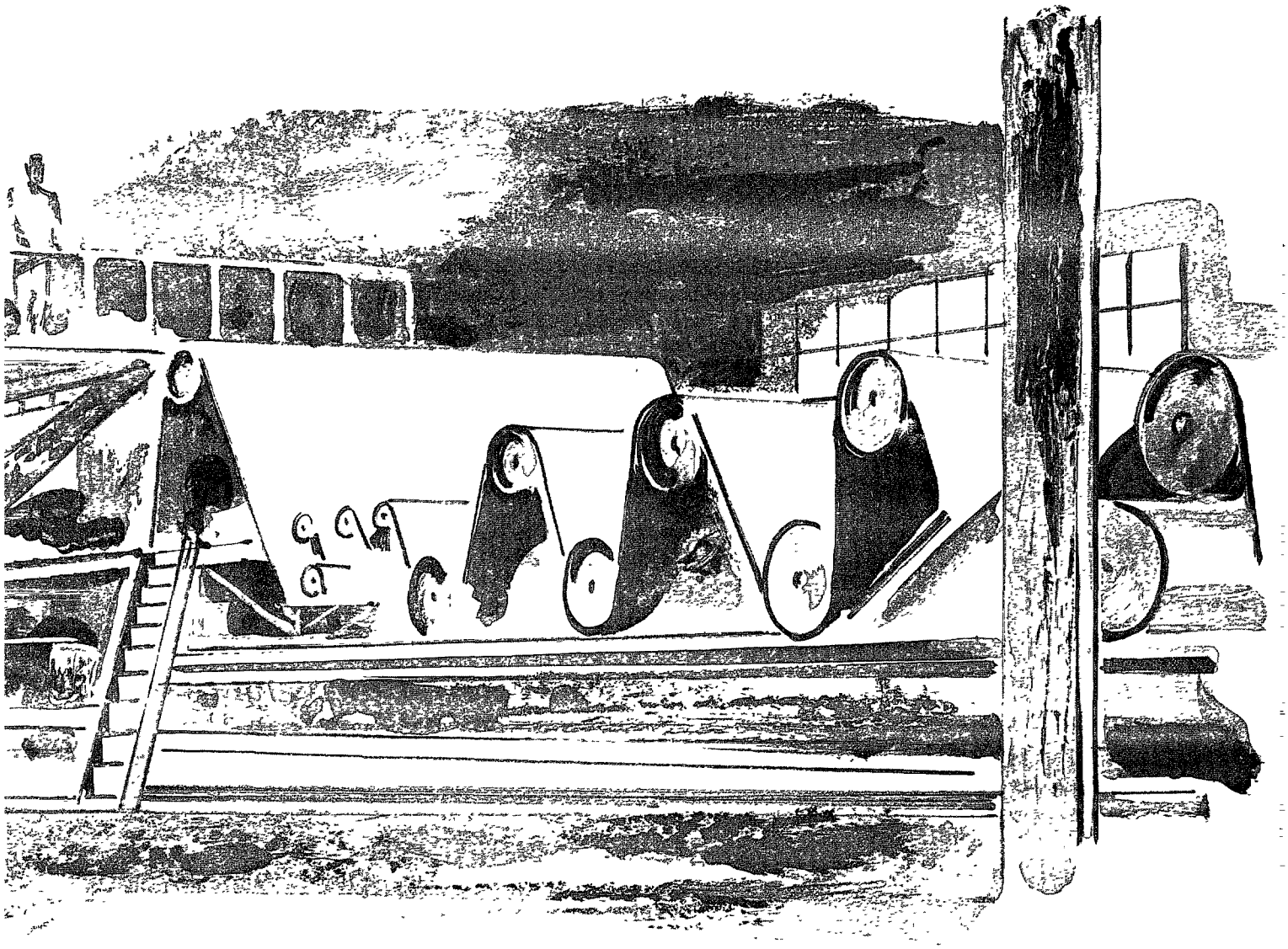


INSTRUCTIONS

GEK-7878

CURRENT-REGULATED HELPER DRIVE CONTROL

FOR MACHINES WITH SILCOMATIC POWER SUPPLIES



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GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

CURRENT-REGULATED HELPER DRIVE CONTROL

INTRODUCTION

These instructions describe the operation, adjustments, and maintenance of the electrical control equipment for current-regulated helper drives. This type control is used where the main-drive power for the machine or section involved is obtained from a lineshaft, a speed-regulated drive, or some mechanical means. The speed of the machine or section involved (and the helper motors) is controlled entirely by the main power source. The current-regulated helper drive simply adds an appropriate torque at the proper place in the machine.

Figure 1 shows, in simplified circuit form, how the various electrical components are connected together as a system. Note that the d-c motor receives a fixed shunt field excitation from the 250-volt d-c constant potential exciter bus. With this arrangement, the amount of torque provided by the motor is approximately proportional to the motor armature current. The armature current is controlled by the regulator amplifier which supplies excitation to the Silcomatic power supply. The Silcomatic output voltage will depend, to a small degree, on the amount of current being circulated, and, to a large degree, on the speed (and therefore CEMF) at which the motor is operating.

The regulator amplifier supplies the Silcomatic excitation necessary to get the Silcomatic voltage to circulate the armature current and also supplies the voltage needed to obtain the motor speed. In other words, the system regulates the armature circuit current (and therefore approximately the motor torque) regardless of motor speed.

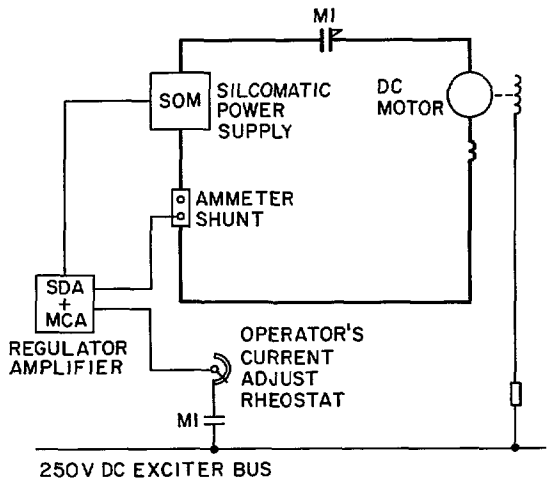


Fig. 1. Simplified circuit diagram of current-regulated helper drive

MAGNETIC CONTROL CIRCUITS

The magnetic control circuit for the current-regulated helper drive will vary depending on the application. Two typical circuits are shown in Figs. 2 and 3.

The circuit shown in Fig. 2 is used when the helper is associated with an electrical speed-regulated drive (such as a couch helper associated with a speed-regulated wire turning roll on an electrical sectional-drive system). In this arrangement, the helper starts and stops with the main (speed-regulated) drive and the start-stop signal is received from an interlock on the main-drive contactor (MDM1). When the main drive is started in the run (or normal) mode, the main-drive run relay (MDRUN) signals the helper to operate in the current-regulated (normal) mode. When the main drive is started in the slow mode, the main-drive run relay (MDRUN) does not pick up and the helper operates in its voltage-regulated (slow) mode. Some cross interlocking of the protective circuits between the helper drive and the main drive is usually provided; the coast-stop relay (CS) interlock going to the main-drive relay circuit is typical of this cross interlocking. Also the interlock labeled SOM in the CS coil circuit is an interlock in the Silcomatic that is closed when the Silcomatic is ready to run. The jumper labeled SDA is removed when the SDA is set up in its "test" mode.

The circuit shown in Fig. 3 is typical of those used when the helper is associated with a lineshaft-driven machine. It works in conjunction with the air clutch control shown in Fig. 3A. The contact labeled SOM in the coast-stop relay (CS) coil circuit is an interlock in the Silcomatic power supply that is closed when the Silcomatic is ready to run. The jumper labeled SDA is removed when the SDA module is set up in its "test" mode. Note that in this particular arrangement, the entire drive is initiated in the run or slow condition by operation of the manual air valve. Whenever the drive is stopped, by action of the STOP pushbutton, turning the selector switch to OFF, or by action of any of the protective devices, the manual air valve must be turned off (to open the air pressure switch and drop out relay R). This resets the lockout relay (LO). The drive may now be restarted.

In some applications the selector switch is replaced by a SLOW pushbutton. In this case the circuits are very similar except that the drive is started in slow by pressing the SLOW pushbutton instead of operating the manual air valve.

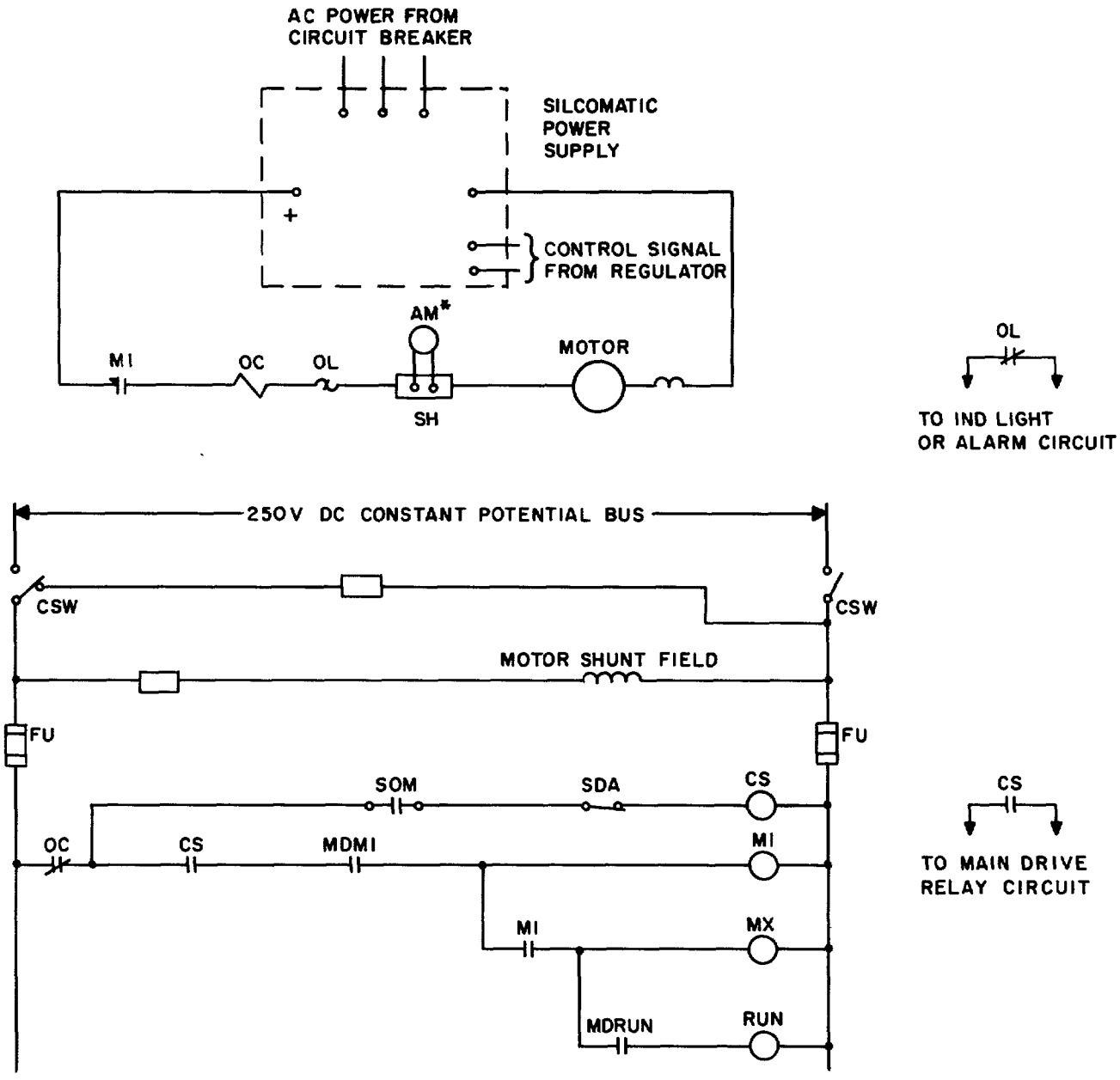


Fig. 2. Typical magnetic control circuit for current-regulated helper drive with electrically-driven main drive

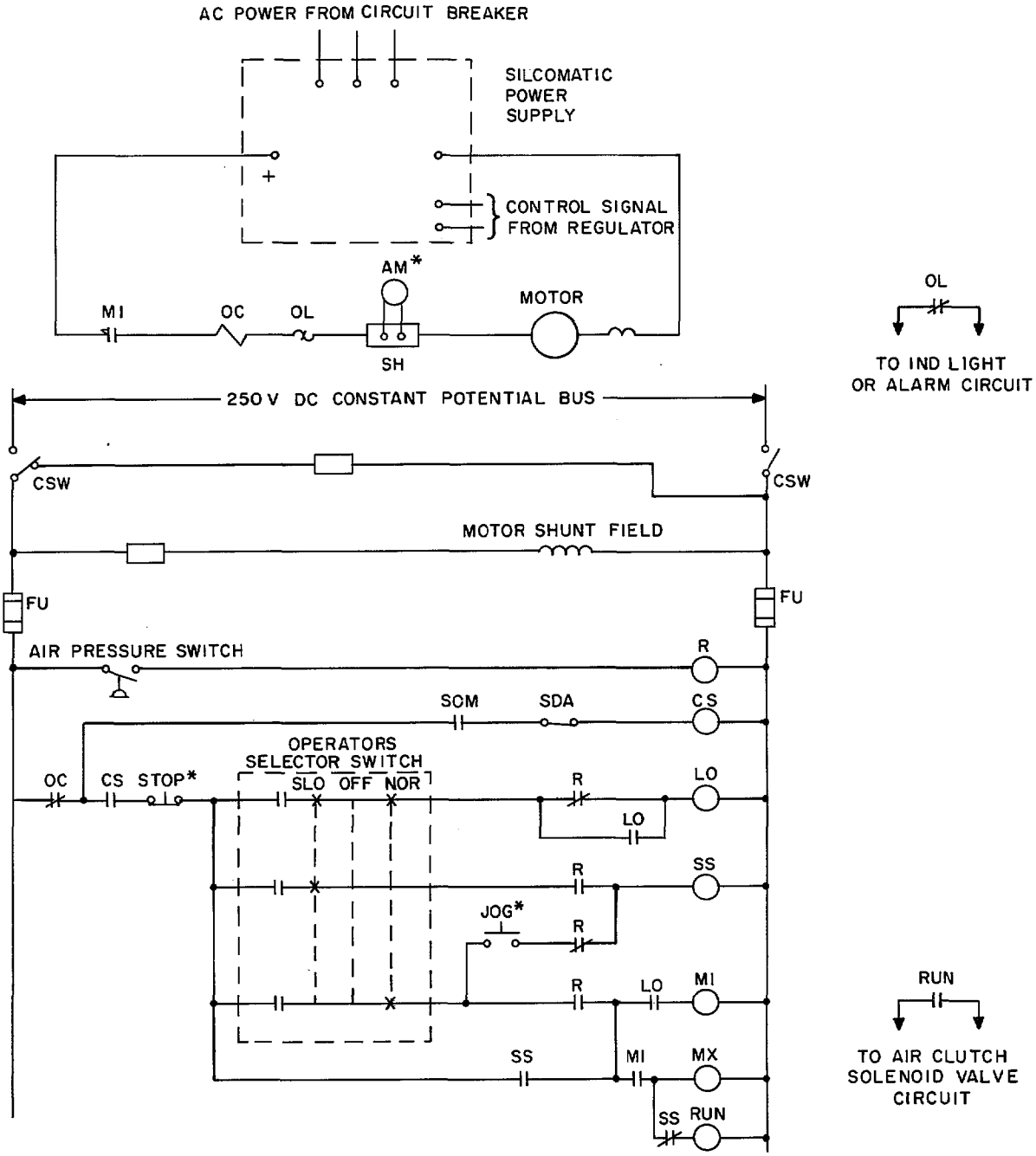


Fig. 3. Typical magnetic control for current-regulated helper drive with lineshaft-driven main drive

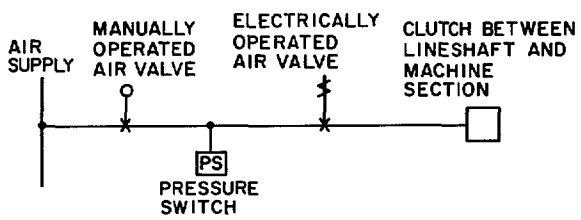


Fig. 3A. Sketch of air-operated clutch control

REGULATOR

GENERAL

Figure 4 shows a one-line diagram for a typical helper drive. The elements labeled MCA and SDA are the regulator modules. MCA is a magnetic amplifier that is used to compare the reference and feedback signals, and SDA is a transistorized driver amplifier for the Silcomatic power supply. These two modules are mounted on the section control panel along with the armature circuit contactor and control relays.

Figure 5 shows a simplified diagram of the basic regulator. The operation of the MCA and SDA modules is explained in the instruction books for those modules (GEI-80353 for MCA and GEI-80348A for SDA), however the following brief explanation of the overall regulator action will help "tie" the various pieces together.

There are three significant "modes" of operation for the system:

1. Standstill (when line contactor is open)
2. Slow operation
3. Normal operation

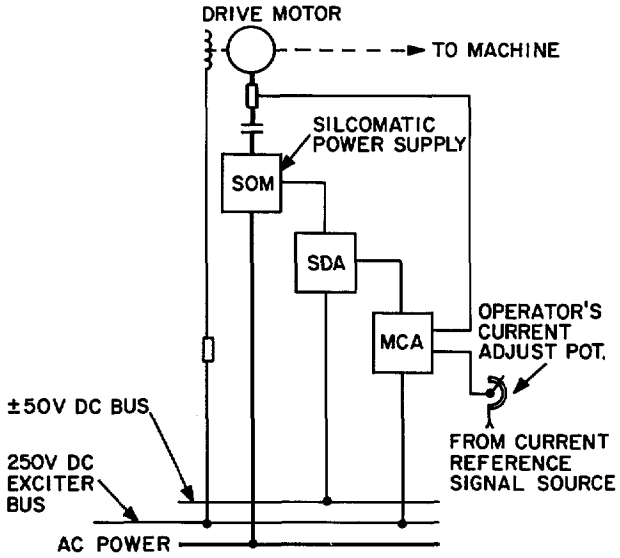


Fig. 4. One-line diagram of typical helper drive

STANDSTILL

When the line contactor is open, the run relay (RUN) and the contactor auxiliary relay (MX) are dropped out. In this mode, the output of the MCA module is shorted-out by a normally closed MX interlock. Thus, a zero reference signal is applied to the SDA operational amplifier through resistor

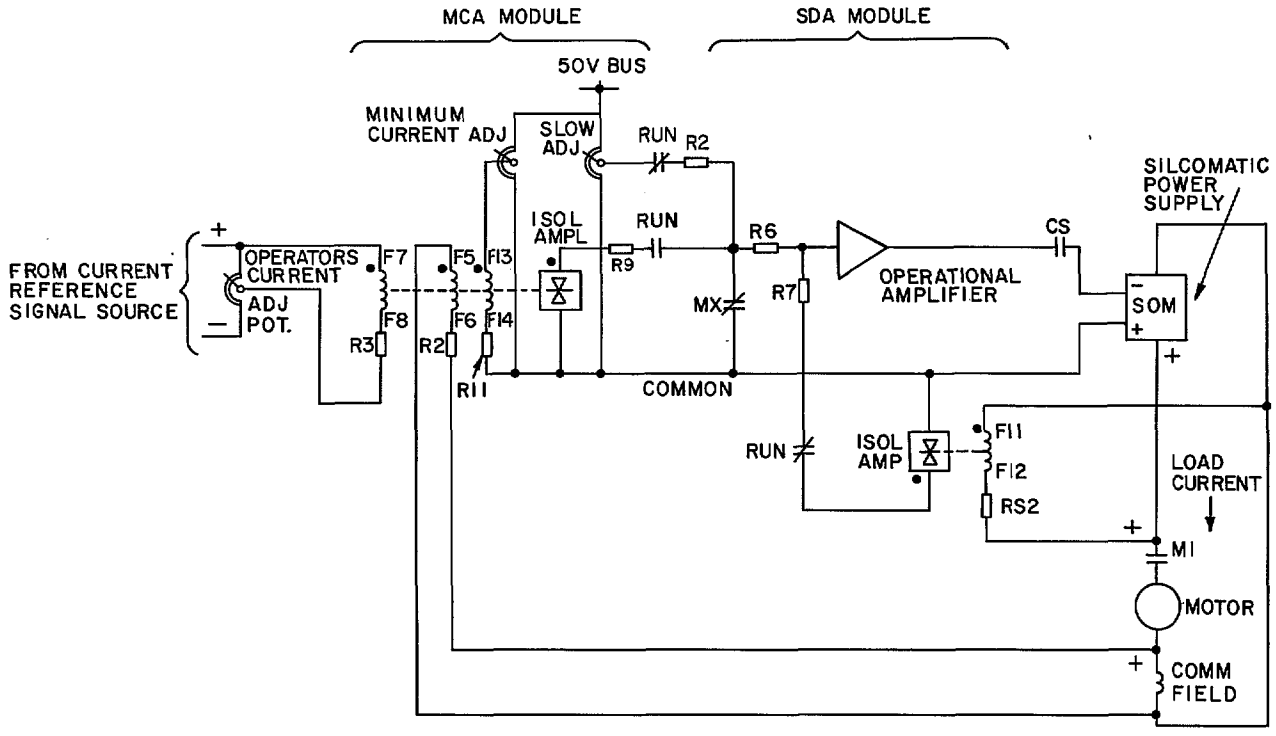


Fig. 5. Simplified circuit diagram of basic regulator

R6. The Silcomatic output voltage is fed back to the SDA operational amplifier via the SDA isolation amplifier, the normally closed RUN contact, and resistor R7. The polarities of the various elements are such that any voltage at the Silcomatic output signals the operational amplifier to apply a voltage to the Silcomatic input that causes the Silcomatic voltage to decrease. The operational amplifier gain is extremely high (approximately 0.007 microamps input current produces full output of 30 volts), so the Silcomatic voltage is held to practically zero volts.

SLOW OPERATION

When the drive is energized in the slow mode, the line contactor (M1) and its auxiliary relay (MX) are picked up but the run relay (RUN) is not picked up. The regulator is now set up so that a reference signal is applied to the SDA operational amplifier from the SLOW ADJ. pot (in the MCA module) via a normally closed RUN contact and resistors R2 and R6. This causes the operational amplifier output to increase (to a negative voltage relative to common) and the Silcomatic voltage to increase. The Silcomatic output voltage is fed back to the SDA operational amplifier via the SDA isolation amplifier, a normally closed RUN contact, and resistor R7. When the Silcomatic output voltage is such that the feedback signal strength matches the reference signal strength, the Silcomatic output voltage remains at that level. Thus in this mode the Silcomatic voltage is regulated to a value proportional to the setting of the SLOW ADJ. potentiometer.

NORMAL OPERATION

When the drive is energized in the normal mode, the line contactor (M1) and the contactor auxiliary relay (MX) and run relay (RUN) are picked up. The regulator is now set up to use both the MCA and the SDA modules. A signal from the current reference signal source is applied to the MCA isolation amplifier winding, F7-F8. This causes the MCA isolation amplifier output to increase, which applies a signal to the SDA operational amplifier (via normally open RUN interlock and resistors R9 and R6) and causes the Silcomatic output to increase. This circulates current in the armature circuit. A voltage proportional to the armature current (typically the motor comm. field IR drop) is fed back to the MCA isolation amplifier winding, F6-F5. When this armature current feedback signal strength matches the current reference signal strength, the regulator is balanced and the armature current remains at that level. Here again, because the gain of the SDA operational amplifier is extremely high, the difference between the reference and feedback signals is very small.

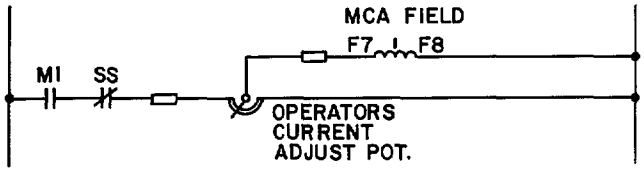


Fig. 6. Basic current reference via operator's pot

In some cases, it is desirable to maintain some minimum value of helper current even when the current reference signal goes to zero (or when the operator turns his Current Adjust Pot to zero). For this purpose, an additional current reference signal can be applied from the minimum current adjust pot via the MCA isolation amplifier winding, F13-F14. This signal is added to the main current reference signal (in the magnetic flux of the MCA isolation amplifier) so that the armature current is regulated to a value proportional to the sum of the two reference signals.

Under certain circumstances, the regulator may tend to oscillate. When this happens, it is necessary to reduce the gain of the system to the point where the oscillations cease. For this purpose, a gain-adjust circuit is included in the MCA module (to keep the diagram simple, this feature is not shown in Fig. 5).

The current reference signal may be supplied by one of the following circuits:

1. From the exciter bus through the operator's current adjust potentiometer as shown in Fig. 6.

2. From the exciter bus through a vacuum-operated potentiometer and the operator's current adjust potentiometer as shown in Fig. 7. In this arrangement, the vacuum-operated rheostat limits the maximum amount of current that the operator can call for.

3. From a GEMAC* pressure to current transducer through the operator's current adjust pot as shown in Fig. 8. The output of the GEMAC trans-

* Registered trade-mark of General Electric Co.

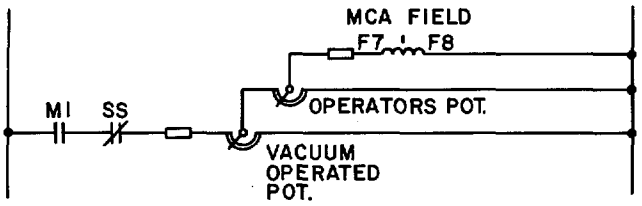


Fig. 7. Current reference via vacuum-operated pot and operator's pot

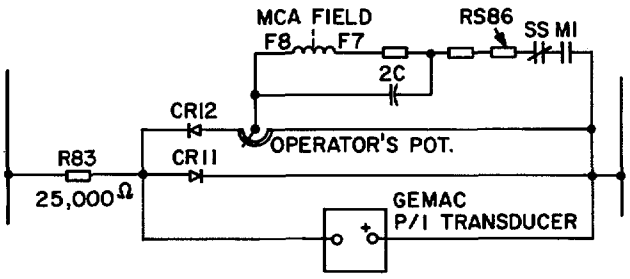


Fig. 8. Current reference via GEMAC transducer and operator's pot

ducer varies from below 10 ma to 50 or more ma as a function of the couch vacuum or main-drive torque. It is desirable to divert the 10 ma minimum GEMAC output from the reference winding. This is done by RS83 and CR11, which form a 10 ma bridge across the 250-volt d-c exciter bus. Until the GEMAC output exceeds 10 ma, the GEMAC output current can be considered to flow backwards through CR11 and practically no current reaches the reference winding. However, when the GEMAC output exceeds 10 ma, the extra current will not be able to flow through CR11 and will therefore flow through the operator's current adjust pot and CR12. The operator's adjust pot can select the desired portion of this GEMAC signal to send to the current reference winding. Another way to look at this is to say that the GEMAC limits the maximum amount of current that the operator can call for.

4. From a signal proportional to the main (or speed-regulated) drive motor current. This arrangement is typically used to control couch helper motor current proportional to wire turning roll motor current, and is shown in Fig. 9. The voltage drop across the wire turning roll motor commutating field is proportional to the wire turning roll motor armature current. It is used as the source of the current reference signal for the couch. The proportion or ratio between the couch and the wire turning motor currents is adjusted by the operator's current adjust potentiometer.

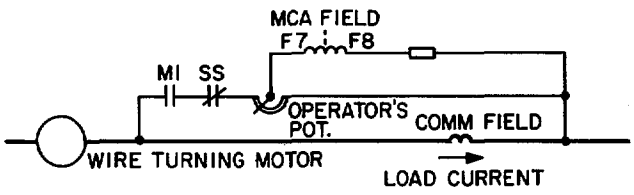


Fig. 9. Current reference via main motor current and operator's pot

INSTALLATION AND TUNE-UP

PRELIMINARY INSPECTION

1. Visually inspect the equipment for shipment damages.

2. If equipment is to be stored before use, provide adequate protection against the elements and other contaminating conditions.

3. If the equipment has been stored or unused for prolonged periods, be sure all equipment is dry and clean before energizing.

4. Manually move all mechanical devices to check for binding or misalignment.

5. Check to assure all interconnecting wires and remote devices agree with the elementary diagram.

6. Check shielded wires to confirm that the shields have been tied down in accordance with the interconnection drawing furnished with equipment.

7. Check terminals for loose connections, electrical clearance, etc.

8. Make sure all contacts are clean and properly positioned.

9. Check motor armatures and fields for proper ohms and grounds. In checking for grounds, avoid subjecting the control equipment to voltages which might cause damage to the semiconductor components.

10. Ensure that the voltage and current feedback signals are being extracted from the proper points on the motors and Silcomatics.

11. Check that adequate electrical, thermal, and mechanical clearances have been maintained.

12. Establish effective communication between control panels and operator's desk, prior to actual tune-up of equipment.

TUNE-UP PROCEDURE

1. General:

Experience has indicated that following certain procedures will result in a more trouble-free and less expensive installation. The following instructions should not be interpreted to represent all conditions which might be encountered or need be considered during any start-up. The actual start-up

procedure should be established by those responsible, giving due regard to safety of personnel and protection of the electrical and mechanical equipments involved. The following instructions assume that the rotating machines and other remote mounted equipment are in proper working order. It is also assumed that the preliminary inspection has been completed and the system is free from grounds, haywires, etc.

2. Initial Conditions:

Before applying power to the control equipment, open the control power disconnect switches and disconnect one motor armature circuit cable to each motor at the control panel. Also disconnect one motor field lead to each motor at the control panel.

3. Exciter and Reference Panel Checkout:

Energize the 250-volt d-c constant potential exciter and check to see that it produces 250 ± 10 volts d-c.

4. Section Panel Magnetic Circuits:

Close the section control power switch and check that the pushbutton, relay, and contactor coil circuits are functioning properly. Use the elementary diagram furnished with equipment to ensure that all circuits are checked.

5. Motor Field Circuits:

Reconnect the motor fields and check to see that the motor field currents are approximately correct. Refer to the elementary diagram machine data sheet furnished with equipment for field current information. Also check that any special field circuits such as field reversal and/or rheostats are operating properly. Final field current adjustments will be made later. Ventilating air should be on if the fields are going to be energized for any period of time.

6. Voltage Regulator:

Set up the SDA module in its "test" mode. That is, move the SDA test card from slot I to slot J. Jumper the CS interlock in the SDA output to the Silcomatic input circuit. Test and set up the SDA as described in the Installation Adjustment section of the SDA instruction book, GEI-80348. Note that the operation of the SDA module in its "test" mode permits manual control (by means of potentiometer RH22 on the SDA) of the SDA output voltage. This feature can be used to check out the operation of the Silcomatic Power Supply. Move the SDA test card back to its normal slot and remove the jumper that was used to short the CS interlock. The Silcomatic output voltage should remain very close to zero volts.

7. Slow-speed Regulator:

Reconnect the motor armature leads and energize the drive in the SLOW mode. Check to see that motor rotation is correct. Adjust the Voltage Reference pot (RH13) in the MCA module to obtain the desired slow operating speed (or for current-regulated couch following the wire turning motor, the desired couch current during Fourdrinier SLOW operation).

8. Current Regulator:

Energize the drive in the normal (current regulated) mode and adjust the current feedback potentiometer (RH10) in the MCA module so that when the operator's current adjust pot is in its maximum current position (and the vacuum operated pot, or GEMAC signal*, or the wire turning motor current** is maximum) the desired maximum regulated helper current is obtained. Then set the operator's rheostat in its zero position and adjust the minimum current adjust pot (RH11) in the MCA module to obtain the desired minimum regulated current. Recheck the maximum current condition again and readjust the current feedback pot (RH10) if necessary.

* **NOTE:** *If GEMAC output current at maximum input to GEMAC P/I transducer is below 60 ma, short out Resistor 86 in Fig. 8.*

** **NOTE:** *If the current signal from the wire turning roll is used as a helper reference, first adjust the current reference signal pot per instructions on the elementary diagram.*

Adjust the gain of the MCA module to obtain maximum current regulator response consistant with stable operation. With the drive running at 30 percent of rated speed (or more) turn the current gain pot (RH12) on the MCA module clockwise until the drive becomes unstable, then turn the pot counterclockwise approximately 20 percent of its travel.

9. Final Motor Field Adjustments

The motor field resistor for the main helper motor should be adjusted so that when the helper is operating fully loaded at full speed the Silcomatic voltage is slightly below rated voltage.

The motor field resistors for the auxiliary helper motors (if supplied) should be adjusted so that when the drive is operating at full speed, and the operator's auxiliary motor field adjust rheo-

stats are at the zero ohm condition, the desired ratio of main to auxiliary motor armature current is obtained.

Remember when making the above adjustments, that the motor field resistance will increase (as much as 25 percent) as the motor temperature increases from room temperature to rated temperature.

10. Armature Dropping Resistors

Where motors of various ratings or like motors operating at varied load settings must operate at reasonably constant load relationships over a wide speed range, it is necessary to have the armature circuit (IR) drop of each motor (at the desired loading) approximately equal to the other motors. In such applications, armature circuit dropping resistors are provided. Determine the load relationships desired and adjust the resistors as required. Use as little resistance as possible in order to minimize the losses. The effect of these resistors is most noticeable at very low speeds, so when checking the load relationships, do so when the machine is at as low a speed as possible.

MAINTENANCE

GENERAL

Periodic inspections and maintenance are important to continuous operation. Regular inspections should be performed so that replacements or repairs can be made before a failure occurs. Costly downtime can be reduced by following three basic rules:

1. Prepare a preventive-maintenance program consisting of items to be checked and recorded at each scheduled inspection. Sometimes frequent inspections are desirable on new equipment. The interval between inspections can then be increased at the user's discretion as he becomes more familiar with the equipment and particular operating conditions.

2. Keep on hand an adequate supply of renewal and repair parts.

3. Much time and trouble can be saved by buying renewal parts from the manufacturer who originally supplied the equipment. Such renewal parts will conform with the specifications for the original parts.

MAINTENANCE

The maintenance schedule and activity outlined in publication GET-1195 "How to Maintain Industrial Control" applies to this control.

RENEWAL PARTS

When ordering renewal parts, specify the complete catalog number of the part. If there is any doubt, include the panel catalog number and the device functional nomenclature.

TROUBLE SHOOTING

During the design and manufacture of this control, everything practical has been done to make it operate consistantly and without failure. However, eventually some trouble may develop. To be prepared for this eventuality, it is suggested that the user:

1. Keep the elementary diagram and instruction books handy.

2. Become familiar with the overall concept of the drive and insofar as possible with the various circuits involved.

3. Keep portable instruments handy for checking voltage, current, and resistance. A portable industrial volt – ohm – milliammeter with a 20,000 ohms per volt d-c sensitivity is the minimum recommended.

4. Keep on hand an adequate supply of spare parts.

Should the drive become inoperative or its performance change, care should be exercised during trouble-shooting in order not to get the drive out of adjustment. Generally, the difficulty stems from a loose connection, broken wire, or failed component and can be corrected without making additional adjustments to the control. Considerable time will be saved if the difficulty can be corrected without changing the component settings necessitating retuning the drive.

The first step in the trouble-shooting procedure should be to identify the problem within a specific area of the system, then systematically reduce this area to a specific module or component. The next step is to repair or replace the faulty component. The final step is to check the drive operation to be sure that no other secondary or contributing problems exist.

The following trouble-shooting guide may be of assistance in tracking down the problem to a general area of the control.

TROUBLE-SHOOTING GUIDE

1. Drive will not start in slow:

   A. Coast-stop relay (CS) not energized.

      1. Check 250-volt d-c exciter bus.

      2. Check protective interlocks in CS coil circuit.

   B. Regulator not operating:

      1. Jumper CS interlock in line between SDA output and Silcomatic input. Set up SDA in its test mode and check operation of SDA and Silcomatic. See SDA instruction book for details. Remove jumper on CS interlock.

   C. General:

      1. Check clutch air pressure and pressure switch (if used).

2. Drive operates in SLOW but not in RUN (or NORMAL):

   A. Line contactor does not pick up. Check relay coil and interlock circuits.

   B. Line contactor picks up but operator has no control of armature current which remains at zero or very low value. Check regulator current reference circuit.

   C. Line contactor picks up and drive immediately shuts down on overcurrent or Silcomatic suppression. Check regulator current feedback circuit.

INDUSTRY CONTROL DEPARTMENT

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